

Table 1: Readout comparison with shared recovered instances. All rows share G1-FC recovery and differ in incumbent readout. A5 retains Native-painted geometry and emits unknown where incumbent FC is unavailable. This is not a pure backbone ablation.

Readout + shared G1	Replica AP	AP50	mIoU	CF18 AP	AP50	mIoU
A2: Native	9.44	23.69	27.96	7.11	15.61	16.44
A5: FC-only	10.94	21.88	23.72	8.32	17.39	17.94
A3: D2	12.39	26.06	30.27	8.23	17.83	19.02

Table 2: Recovery runtime with the same measured return boundary. Host stages are exclusive and sum to total wall time. GPU events are non-additive elapsed diagnostics, not extra wall time. U2 has an archived observation policy; G1 performs fresh visibility search.

Stage (s)	U2 control	G1 reference	G1 selected
registry geometry	0.677	2.426	2.060
view search	0.498	34.922	9.780
rgb preprocess	0.120	0.445	0.357
recognition	0.153	0.560	0.561
export bookkeeping	4.508	6.070	6.071
other sync	0.007	0.024	0.024
complete incremental total	5.962	44.448	18.853
GPU event diagnostics (non-additive)			
encoder	0.132	0.476	0.476
region pool head	0.005	0.022	0.022